

1. Introduction

Accurately predicting Airbnb listing prices is an important challenge in the short-term rental market, where hosts must set competitive yet profitable prices. This project applies machine-learning regression techniques to forecast listing prices using publicly available Airbnb Open Data. The dataset contains **102,352 listings and 25 columns**, representing a diverse mix of numerical and categorical attributes across New York City boroughs ().

The workflow follows a standard machine-learning pipeline: **data preprocessing, model training, ensemble learning**, and **evaluation**. The primary objective is to identify the model that delivers the most accurate and stable price predictions.

2. Dataset Overview

The dataset includes **24 predictive features** describing listing characteristics such as location, room type, host attributes, and availability. The **target variable** is *price*. The price distribution is **highly skewed**, with notable outliers—common in real estate and rental markets. Several columns contain missing values, particularly *license* and *house_rules*, requiring careful preprocessing ().

3. Preprocessing

To prepare the data for modeling:

- **Rows with missing price values were removed**, ensuring the target variable is complete.
- **Identifier columns** (*id*, *host_id*) were dropped because they do not contribute predictive value.
- **Categorical features (14 total)** were transformed using **One-Hot Encoding**, expanding them into binary indicator variables.
- **Numeric features** were standardized using **StandardScaler** to improve model stability.
- The dataset was split into **70% training**, **15% validation**, and **15% testing**, enabling reliable model selection and unbiased final evaluation ().

4. Models Implemented

A diverse set of regression algorithms were trained to compare performance across linear, tree-based, and ensemble approaches:

- **Linear Regression**
- **Decision Tree**
- **Random Forest**